

AtmoSync Micro-Climate Analytics

Business Insights & Recommendations Report

This report summarizes findings from 50,000 IoT sensor readings across 550 refrigerated shipping containers transporting perishable commodities (avocado, banana, blueberry, grapes, mango, strawberry, tomato) from 7 origin ports to primary and secondary markets. It translates the dashboard's data into clear operational and commercial recommendations.

Key Metrics at a Glance

Metric	Value
Total sensor readings	50,000
Total containers tracked	550
Average product quality score	95.6 / 100
Readings with significant temperature drift	5.9%
Containers with faulty cooling units	102 of 550 (18.5%)
Total revenue impact of rerouting (all commodities)	-\$234.9M
Average shelf life remaining at last reading	1,600 hours (~66.7 days)

Insight 1: Rerouting Shipments Is a Loss-Prevention Move, Not a Profit Strategy

Across all 50,000 readings, the total revenue impact of rerouting shipments from primary to secondary markets is -\$234.9 million. This is because secondary market prices are consistently lower than primary market prices for the same commodity. Rerouting is only recommended for containers already flagged At-Risk or Critical (1,617 readings, 3.2% of the dataset) — in those cases, accepting a lower price is preferable to losing the shipment entirely to spoilage.

Blueberry (-\$63.3M) and Strawberry (-\$44.4M) account for the two largest revenue losses from rerouting, likely due to their higher price sensitivity and shorter shelf life compared to hardier commodities like Banana (-\$13.4M).

What the company should do

- Treat rerouting cost as an accepted insurance expense against total spoilage loss, not a KPI to minimize on its own.
- Prioritize investment in temperature control for Blueberry and Strawberry shipments specifically, since they carry the highest financial exposure when rerouting is needed.

- Track "rerouting cost avoided spoilage value" as a paired metric, so the -\$234.9M is understood alongside the (larger) value of product saved from total loss.

What the company should not do

- Do not cut back on rerouting as a cost-saving measure — this would increase full-spoilage losses, which are typically worse than a discounted sale.
- Do not treat all commodities the same when budgeting for this cost — Blueberry and Strawberry need disproportionate attention relative to Banana or Mango.

Insight 2: Tangier Port Has the Highest Risk Concentration

When comparing At-Risk and Critical reading rates across origin ports, Tangier Port (MA) stands out with 4.4% of its readings flagged At-Risk or Critical — the highest of all 7 ports — followed by Guayaquil Port (3.6%) and Chennai Port (3.4%). Callao Port (PE) has the lowest risk rate at 2.2%.

What the company should do

- Conduct a targeted equipment and process audit on containers departing from Tangier Port to identify root causes (equipment age, handling delays, route duration).
- Use Callao Port's lower risk rate as a benchmark — compare procedures between Callao and Tangier to identify transferable best practices.
- Consider adding stricter pre-departure cooling checks specifically for Tangier-origin shipments.

What the company should not do

- Do not assume the difference is random — a 2x gap between the highest and lowest-risk ports (4.4% vs 2.2%) is large enough to warrant investigation rather than dismissal.
- Do not apply a blanket fix across all ports — the cause at Tangier may not be the same issue affecting other ports, if any.

Insight 3: Temperature Deviation Directly Damages Product Quality

There is a clear, direct relationship between temperature deviation and product quality score. Readings within 1°C of the ideal range average a 97.9 quality score, but this drops sharply as deviation increases — down to 84.5 at 2-3°C deviation, and just 46.0 for readings with more than 5°C deviation from the ideal range.

Temperature Deviation	Average Quality Score
0 - 1°C	97.9
1 - 2°C	92.0
2 - 3°C	84.5
3 - 5°C	68.7
5°C or more	46.0

What the company should do

- Set an internal alert threshold at 2°C deviation (not just at the 3°C "significant drift" flag), since quality already drops noticeably (to 84.5) before the current flag triggers.
- Use this table to financially justify cooling equipment upgrades — a container drifting past 5°C is losing more than half its quality score on average.
- Prioritize real-time alerts for any container crossing 2°C deviation so corrective action (rerouting, faster delivery) can happen before quality drops further.

What the company should not do

- Do not rely solely on the current 3°C significant-drift flag as the only warning signal — meaningful quality loss is already happening before that threshold.
- Do not treat minor deviations (under 1°C) as a concern requiring action — quality remains high (97.9) in that range, so resources are better spent on higher-deviation cases.

Insight 4: Cooling Equipment Reliability Is a Widespread Issue

102 of 550 containers (18.5%) recorded a faulty cooling unit at some point during transit. This is a large enough share of the fleet to be a systemic maintenance issue rather than isolated incidents.

What the company should do

- Introduce a scheduled preventive maintenance program for cooling units rather than relying on reactive repairs after failure.
- Cross-reference the 102 faulty-cooling containers against the risk status data to confirm equipment failure is a leading driver of At-Risk/Critical readings.
- Track cooling unit fault rate as an ongoing operational KPI, not just a one-time data point.

What the company should not do

- Do not treat this as a minor issue — nearly 1 in 5 containers experiencing a cooling fault is a fleet-wide reliability problem, not a few bad units.
- Do not delay maintenance investment while treating rerouting/discounting as the only mitigation — fixing the equipment addresses the root cause rather than just managing the symptom.

Prepared from the AtmoSync cleaned dataset (50,000 readings, 550 containers) as part of a data analyst internship project. All figures are derived directly from the underlying data.